

Model-form versus finite-sampling uncertainty in micromagnet-assisted spin–valley shuttling: a periodic-reference sensitivity atlas

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Coherent electron spin shuttling in Si/SiGe couples spin to the valley degree of freedom, so design studies of micromagnet geometry must assume a local spin–valley coupling profile $\lambda_{sv}(x)$ that is not independently fixed. We ask how far design-facing conclusions depend on that assumption, and how that dependence compares with the variation a finite Monte-Carlo ensemble produces on its own. Using a periodic micromagnet array as a controlled reference geometry, we evaluate a 108-condition grid under four diagnostic coupling ansätze. Two comparisons are placed on one scale: the *cross-ansatz* difference, obtained by changing the coupling model on a fixed set of noise realizations, and the *cross-seed* difference, obtained by changing the noise sample under a fixed coupling model. Disagreement is the fraction of conditions the two members of a pair assign to different nine-category labels, formed by classifying each of the two response channels as improve, inactive, or worsen. Cross-ansatz disagreement is 0.388 at the pre-registered endpoint of $n = 30$ realizations and 0.395 at the registered follow-up endpoint of $n = 100$, and stays within a three-point band across all eight checkpoints, while same-ansatz cross-seed disagreement falls monotonically from 0.542 to 0.298. At the registered checkpoints through $n = 30$ the sampling disagreement is comparable to or larger than the cross-ansatz disagreement; from $n = 40$ to 100 the ordering is reversed. The balance is observable dependent: resolving the label into its two response channels, the cross-ansatz to cross-seed ratio at $n = 100$ is 5.60 for spin dephasing and 0.96 for valley excitation, a post-hoc decomposition. A post-hoc ranking diagnostic shows that rank correlation across ansätze does not track the classification: two ansatz pairs with nearly identical rank correlations differ by 37 percentage points in classification agreement. Operating points selected for stability across noise samples alone remain robust under all four coupling profiles in some cases and under none in others. Design conclusions obtained under one assumed coupling profile therefore remain model-conditional until the effective coupling is independently constrained, or robustness is demonstrated across a justified model family.

I. INTRODUCTION

Coherent transport of electron spin qubits between distant sites (“shuttling”) lets qubit registers sit apart from readout and control hardware, and is a key enabler for scalable Si/SiGe architectures^{1–6}. The platform is mature: high-fidelity gates, multi-qubit processors, error correction, established single-spin readout and an industrial fabrication path are all in place^{7–20}. Conveyor-mode and bucket-brigade shuttling have been demonstrated at high fidelity^{21–23}, as have shuttling-based two-qubit links between separated registers^{24–26}, and gates acting on moving spins have been proposed theoretically²⁷. A central obstacle is the valley degree of freedom. The two low-lying conduction-band valleys in strained Si are nearly degenerate, so a shuttled electron can leak into the excited valley. Spatial variation of the valley energies along the path therefore couples motion to the valley sector. Experimentally, low- E_{VS} pockets arising from alloy disorder are a known threat to shuttling fidelity^{28–30}, and recent shuttling-based maps report physical valley splittings E_{VS} from 1.5 to 200 μeV along real conveyor trajectories, dominated by alloy disorder, with trajectory selection around low-splitting regions demonstrated as a prac-

tical mitigation³¹. When combined with a micromagnet-induced synthetic spin–orbit field^{32,33}, valley transitions become correlated with spin dephasing, so that geometry and trajectory choices affect spin coherence and valley leakage simultaneously.

A natural program is *valley-aware co-design*: jointly optimizing micromagnet geometry and shuttling trajectory to suppress both valley leakage and dephasing. Any such optimization assumes a model of how the two sectors couple—an effective spin–valley Hamiltonian H_{sv} . The literature motivates several ingredients, among them gradient-induced synthetic spin–orbit coupling, valley-pocket localization, and spin–valley hot spots, but leaves the local profile $\lambda_{sv}(x)$ for a shuttled dot open.

We ask a prior question: *how much do co-design conclusions depend on that choice, and how does that dependence compare with the variation a finite Monte-Carlo ensemble produces on its own?* The comparison is the point. Showing that four coupling profiles give four answers means little if the same simulation gives different answers to itself.

We therefore fix the geometry, the operator structure, the noise model and the condition grid, and drive all four ansätze with common random numbers, so that every cross-ansatz comparison is made on one realization set. Against it we place the disagreement measured between independent seed blocks under a fixed ansatz. Both are counted on the same design-facing classification: whether

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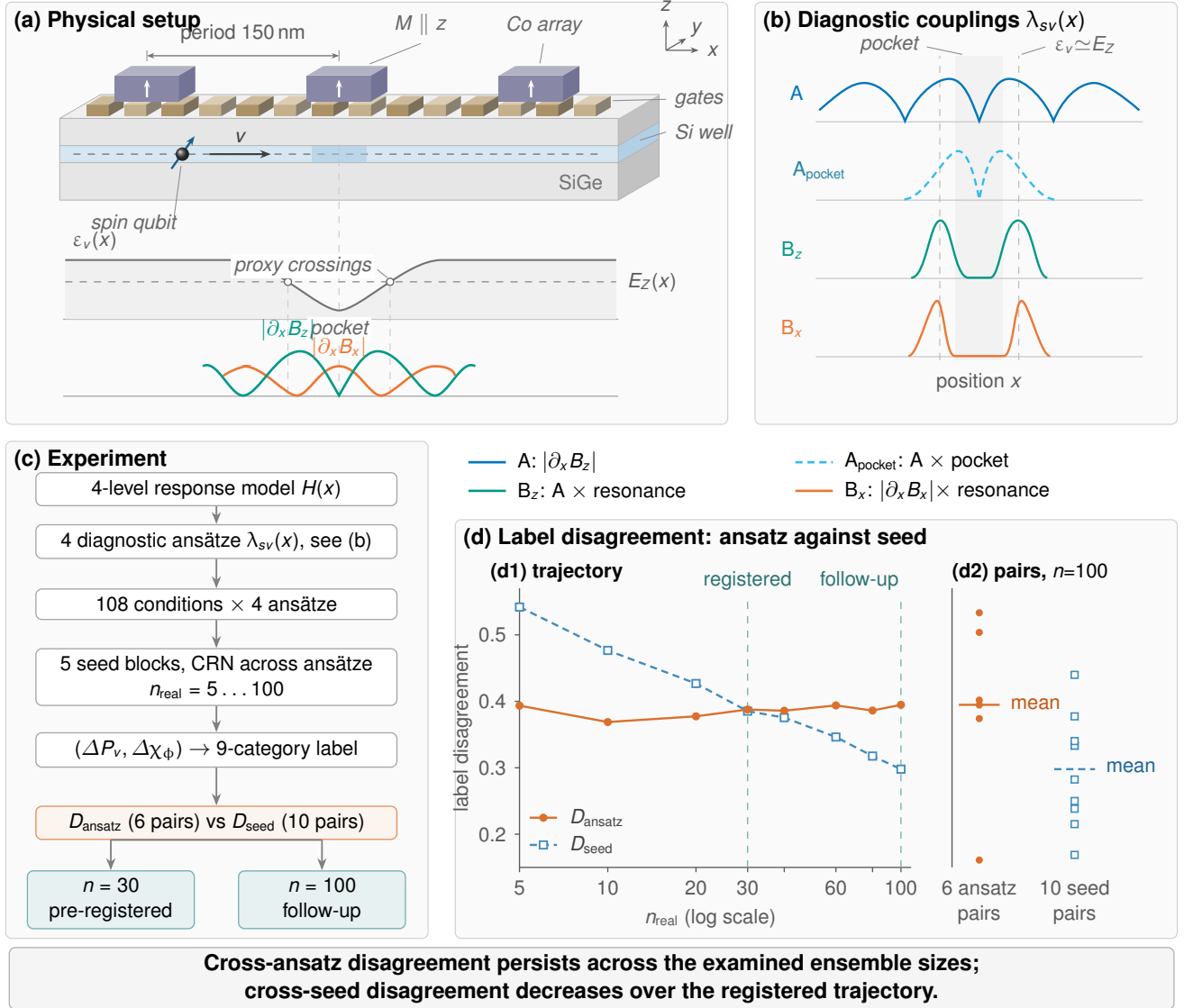


FIG. 1. Overview. (a) Device schematic: an electron spin (tilted arrow) is shuttled at constant velocity v through the Si quantum well of a Si/SiGe heterostructure, beneath a periodic array of identical, uniformly z -magnetized rectangular Co prisms (period 150 nm; not to scale along y), a controlled reference geometry motivated by proposed shuttling-bus architectures. Below, aligned with the device: the diabatic valley-detuning landscape $\epsilon_v(x)$ of Eq. (1), the input-detuning proxy crossings where $\epsilon_v(x) \simeq E_Z(x) = g\mu_B[B_{\text{ext}} + B_z(x)]$ (circles), and the gradient components $|\partial_x B_z|$ and $|\partial_x B_x|$, which are approximately phase shifted in space. At the array-symmetry point $x = 0$ the former vanishes and the latter is maximal; the centred-pocket case places x_c there, so the node-pocket coincidence follows from that placement and is not forced (Sec. III). E_Z is drawn as a horizontal level for clarity; the calculations use its full spatial dependence (Fig. 2). (b) The four coupling ansätze $\lambda_{sv}(x)$ on a common position axis; the grey band marks the pocket and the dashed lines the proxy crossings. For a centred pocket the resonance ansätze (B_z , B_x) activate symmetrically on both flanks, with B_x peaking slightly outboard of the $\partial_x B_z$ -based kernels on the same flanks. Curves are independently height-normalized; peak positions are those of the released profiles. The two-line legend below the upper row gives mnemonic descriptors for the four profiles; the kernels themselves are defined in Sec. III. (c) Experiment: 108 conditions, each evaluated under all four ansätze in each of five seed blocks, with common random numbers shared across ansätze within a block, at nested realization counts $n_{\text{real}} = 5 \dots 100$. Each condition receives a nine-category thresholded label. Cross-ansatz disagreement D_{ansatz} , averaged over the six ansatz pairs, is measured against same-ansatz cross-seed disagreement D_{seed} , averaged over the ten seed-block pairs. $n_{\text{real}} = 30$ is the pre-registered endpoint; $n_{\text{real}} = 100$ is the endpoint of a follow-up extension registered after the $n_{\text{real}} = 30$ result was seen. (d) Disagreement is the mean fraction of conditions assigned different labels by the two members of a pair. (d1) Across the eight registered checkpoints D_{seed} falls from 0.542 to 0.298 while D_{ansatz} stays within 0.369–0.395; the ordering reverses between $n_{\text{real}} = 20$ and 30. (d2) The underlying pairs at $n_{\text{real}} = 100$ on the same vertical scale. The two distributions overlap, the seed range 0.169–0.440 lying inside the ansatz range 0.161–0.533, so uniform dominance does not hold.

each of the two coupling-induced responses is negative, positive, or below its operational floor, against floors fixed in advance. We repeat the comparison at Monte-Carlo ensemble sizes from $n_{\text{real}} = 5$ to 100, where n_{real} is the number of independent charge-noise realizations averaged per condition. Figure 1 summarizes the setup, the four ansätze, the experiment, and the matched comparison.

The estimands, the candidate-selection rule and the checkpoints $n_{\text{real}} = 5, 10, 20, 30$ with endpoint $n_{\text{real}} = 30$ were fixed in writing before any of these data existed. After the $n_{\text{real}} = 30$ result showed cross-seed disagreement still falling, we registered a follow-up extension to $n_{\text{real}} = 100$, with checkpoints, hard stop and reporting language fixed before the extension data were generated. Diagnostics added after seeing results are marked post-hoc where they appear.

An earlier archived calculation of this atlas drew a separate noise realization for each ansatz, so a cross-ansatz difference there also carried a sampling difference. We therefore begin by quantifying how much that confounded the apparent model disagreement, before turning to the matched comparison.

The main findings are as follows. Cross-ansatz disagreement — the fraction of conditions two ansätze label differently — is 0.395 at $n_{\text{real}} = 100$ and stays within a three-point band across all eight checkpoints, while cross-seed disagreement falls monotonically from 0.542 to 0.298; at the registered checkpoints through $n_{\text{real}} = 30$ the sampling disagreement is comparable to or larger than the cross-ansatz disagreement, and at the checkpoints from 40 to 100 the ordering is reversed. The balance is observable dependent: resolving the label into its two response channels, the cross-ansatz to cross-seed ratio at $n_{\text{real}} = 100$ is 5.60 for spin dephasing and 0.96 for valley excitation, a post-hoc decomposition. A post-hoc ranking diagnostic shows that rank correlation across ansätze does not track the classification: two ansatz pairs whose rank correlations differ by 0.011 differ by 37 percentage points in classification agreement. Operating points selected for stability across noise samples alone remain robust under all four coupling profiles in some cases and under none in others.

II. RELATED WORK AND POSITIONING

Prior studies of valley physics in Si/SiGe have characterized valley splitting statistics^{28,34}, the role of interface disorder and valley-orbit coupling^{35,36}, and shuttling-induced valley excitation^{2,29,37,38}. Work on micromagnet-based spin manipulation^{32,33} has established synthetic spin-orbit coupling as a control resource, and the role of spin-orbit interaction and g -tensor anisotropy in quantum dots has been studied in both electron and hole systems^{39–42}. Charge-noise dephasing during shuttling and in static dots has been modeled and measured with filter-function and disorder-aware ap-

proaches^{2,18,43–45}, and hopping/baseband control with site-dependent quantization axes has emerged as an alternative to resonant driving^{32,46}. Disorder and strain effects on spin-orbit coupling have been quantified in related Ge hole-spin device models^{47,48}. Trajectory and device parameters for shuttling *have* been optimized in specific studies, including velocity and design trade-offs in blueprint analyses and disorder-aware shuttling strategies^{2,29}. Generic optimal-control machinery is also available^{49,50}.

In such studies the effective coupling model is held fixed, which is the right choice where the model is validated for the regime in question. The setting we address is the local spin-valley coupling of a dot shuttled through a micromagnet field. No generally validated local $\lambda_{sv}(x)$ that can be adopted without additional modeling assumptions is yet available for this setting (Sec. III), so there the model choice itself becomes a source of design risk. Quantifying that risk is the purpose of this work. Recent conveyor-mode measurements increasingly constrain the relevant microscopic ingredients. They provide spatial maps of the valley splitting E_{VS} along shuttling trajectories and, together with valley-resolved g -factor maps, constrain the complex intervalley coupling^{31,51}. These measurements do not by themselves determine the effective local moving-dot spin-valley kernel parameterized here. They sharpen the experimental target that a future microscopic $\lambda_{sv}(x)$ must fit. Valley-aware co-design in the sense used here means jointly tuning geometry and trajectory against both valley leakage and dephasing. We accordingly treat it as a prospective program.

Our contribution is methodological. We fix the four-level response model, the reference geometry and the condition grid, and vary two things against each other: the coupling ansatz, under common random numbers, and the seed block, at fixed ansatz. Both are tracked across ensemble size. Each ansatz is a kernel multiplying one fixed minimal spin-valley operator. Micromagnets and electron shuttling are individually established experimentally^{21,22,32,33}, but periodically patterned micromagnet shuttling architectures remain largely proposed or emerging designs^{1,2,32}; we accordingly treat the periodic array as a controlled reference geometry for the diagnostic comparison. The design conclusions drawn on that geometry remain conditional on the assumed coupling profile. They can be strengthened by independently constraining that profile, or by demonstrating robustness across a justified model family.

III. MODEL

One part of the simulator varies in this study: the spatial form of the spin-valley coupling, $\lambda_{sv}(x)$. The geometry, the noise model, the operator structure and every other model ingredient are held fixed. The sections below set up that fixed backdrop and then give the four

TABLE I. Representative simulation parameters and their physical motivation. Values are chosen to be representative of Si/SiGe micromagnet-shuttling devices; the study varies geometry and trajectory around these baselines; no value is fitted to a specific device. The valley-energy scales are motivated by measured Si/SiGe valley splittings E_{VS} ^{28,34} and by conveyor-trajectory maps^{31,37}. Literature citations identify the experimental or design scales that motivate each choice; the individual entries are not copied from the cited devices, and the reference geometry is checked independently against micromagnet field and gradient scales in the Supplementary Material. The simulator’s numerical consistency and its reproduction of the relevant literature limits are documented in the Supplementary Material.

parameter	value	motivation / representative scale
micromagnet period	150 nm	controlled-reference period (see caption)
half_x/half_z	25/15 nm	$50 \times 50 \times 30$ nm reference prism
saturation M_s	1.4×10^6 A/m	Co-like model value
external field B_{ext}	0.5 T	chosen operating-field baseline; enters Eq. (3)
$\varepsilon_{v,\text{base}}$ far from pocket	100 μeV	baseline motivated by measured E_{VS} scales
$\varepsilon_{v,\text{min}}$ at pocket centre	5 μeV	low-detuning floor within measured E_{VS} scales
pocket width σ_p	30 nm	sets path span $8\sigma_p$
valley off-diagonal Δ_v	0.5 μeV	avoided-crossing scale (toy)
resonance width σ_E	10 μeV	resonance-localization envelope width
shuttling velocity v	5–20 m/s	conveyor-mode sweep ^{2,22}

alternative forms of $\lambda_{sv}(x)$.

We consider an electron shuttled at constant velocity v along x through a periodic micromagnet array, with a position-dependent diabatic valley detuning

$$\varepsilon_v(x) = \varepsilon_{v,\text{base}} - (\varepsilon_{v,\text{base}} - \varepsilon_{v,\text{min}}) \exp\left[-\frac{(x - x_c)^2}{2\sigma_p^2}\right], \quad (1)$$

a baseline with a Gaussian pocket (minimum) near the shuttling path, using $\varepsilon_{v,\text{base}} = 100 \mu\text{eV}$, $\varepsilon_{v,\text{min}} = 5 \mu\text{eV}$ and $\sigma_p = 30$ nm; for A_{pocket} we set $\sigma_\lambda = \sigma_p$. The constant-velocity trajectory spans $x \in [x_c - 4\sigma_p, x_c + 4\sigma_p]$, a total path length of 240 nm. This detuning landscape is a phenomenological surrogate for the measured spatial variation of the physical valley splitting E_{VS} . That variation is expected from alloy-disorder and interface physics^{2,28,34}, and has been mapped directly along a shuttling path by conveyor-mode spin-coherent measurements³⁷. Representative values for the device geometry, valley energetics, and shuttling velocity are summarized in Table I.

We model the spin and valley degrees of freedom with a minimal four-level effective Hamiltonian

$$H(x) = \frac{1}{2}g\mu_B[B_{\text{ext}} + B_z(x)]\sigma_z \otimes I_v + \frac{1}{2}g\mu_B B_x(x)\sigma_x \otimes I_v - \frac{1}{2}\varepsilon_v(x)I_s \otimes \tau_z + \Delta_v I_s \otimes \tau_x + \lambda_{sv}(x)\sigma_x \otimes \tau_x, \quad (2)$$

where σ_i and τ_i are Pauli operators on the spin and valley two-level spaces, $B_z(x)$ and $B_x(x)$ are the longitudinal and transverse components of the *micromagnet stray field*, and $B_{\text{ext}} = 0.5$ T is the uniform external field along z . The spin block is the Zeeman Hamiltonian in the total longitudinal field $B_{\text{ext}} + B_z(x)$ together with the micromagnet transverse field; the transverse term $g\mu_B B_x(x)$ is a position-dependent transverse Zeeman *energy*, not an AC EDSR drive. Because B_{ext} is spatially uniform it drops out of every spatial derivative, so the

gradient-based ansätze below use the micromagnet stray-field gradients $\partial_x B_z(x)$ and $\partial_x B_x(x)$. The valley block is a generic two-level avoided-crossing model with diabatic detuning $\varepsilon_v(x)$ and off-diagonal coupling Δ_v , so its isolated eigenlevel splitting is $E_{\text{gap}}(x) = \sqrt{\varepsilon_v(x)^2 + 4\Delta_v^2}$. Over the 5–100 μeV input range used here this exceeds $\varepsilon_v(x)$ by less than 2%, with the largest relative difference at the pocket minimum and less than 0.02% near the $\varepsilon_v \simeq E_Z$ crossings that set the proxy hot spots. For the resonance kernels the consequence is smaller still. The exact dressed degeneracy condition is $E_{\text{gap}} = E_{\text{spin}}$; centring the window on $\varepsilon_v \simeq E_Z$ instead moves the crossing by 0.0014 μeV , 0.014% of the $\sigma_E = 10 \mu\text{eV}$ window width, because the two approximations displace it in the same direction and largely cancel. That cancellation is a consequence of $\Delta_v \ll \varepsilon_v$: at $\Delta_v = 5 \mu\text{eV}$ the residual would reach 8% of the window width.

Three valley energies appear here: the diabatic input detuning $\varepsilon_v(x)$, the valley-block eigenlevel splitting $E_{\text{gap}}(x)$, and the physical valley splitting E_{VS} reported by experiment. Of the two model quantities, E_{gap} is the analogue of a measured E_{VS} . The measured E_{VS} range motivates the scale chosen for ε_v ; for $\Delta_v = 0.5 \mu\text{eV}$, E_{gap} and ε_v agree to within 2% over that range. Δ_v is a phenomenological mixing parameter; microscopically such intervalley coupling arises from interface and valley-orbit physics³⁵.

This Hamiltonian is a diagnostic effective model, not a microscopically derived Si/SiGe H_{sv} . The term $\lambda_{sv}\sigma_x \otimes \tau_x$ is a minimal real spin-flip/valley-flip Pauli product, one of several possible structures (e.g. $\sigma_x\tau_y$ or $\sigma_y\tau_x$). Microscopically, spin–valley mixing follows from matrix elements of intrinsic and synthetic spin–orbit terms between valley-dependent envelope states^{36,52}. All four ansätze share this one operator structure and differ only in the scalar profile $\lambda_{sv}(x)$: four spatial kernels on one fixed operator, not four microscopic Hamiltonians. That scope holds for every statement about the ansätze below.

The coupling strength is set by a profile $\lambda_{sv}(x)$ whose form is itself uncertain, so we compare four toy-level ansätze. Each is motivated by a known ingredient without being derived from it: spatial Zeeman-energy variation from the micromagnet gradients^{2,43}; transverse-gradient synthetic spin-orbit coupling^{36,53,54}; valley-pocket localization^{28,29,37}; and the spin-valley hot spot at near-degeneracy of the dressed spin and valley sectors^{52,55}, which the released kernels represent by the input-detuning proxy $\varepsilon_v(x) \simeq E_Z(x)$. We use the physically transparent total-field convention throughout, in which the Zeeman energy is evaluated with the full local field,

$$E_Z(x) = g\mu_B [B_{\text{ext}} + B_z(x)], \quad (3)$$

with $B_{\text{ext}} = 0.5 \text{ T}$ (Table I). The four ansätze are then

$$\text{A: } \lambda_{sv}(x) = \lambda_0 \frac{|\partial_x B_z(x)|}{\max_x |\partial_x B_z|}, \quad (4)$$

$$\text{A}_{\text{pocket:}} \lambda_{sv}(x) = \lambda_{sv}^{\text{A}}(x) \exp\left[-\frac{(x - x_c)^2}{2\sigma_\lambda^2}\right], \quad (5)$$

$$\text{B}_z: \lambda_{sv}(x) = \lambda_{sv}^{\text{A}}(x) \exp\left[-\frac{(\varepsilon_v(x) - E_Z(x))^2}{2\sigma_E^2}\right], \quad (6)$$

$$\text{B}_x: \lambda_{sv}(x) = \lambda_0 \frac{|\partial_x B_x(x)|}{\max_x |\partial_x B_x|} \exp\left[-\frac{(\varepsilon_v(x) - E_Z(x))^2}{2\sigma_E^2}\right], \quad (7)$$

where $E_Z(x)$ is the local Zeeman energy of Eq. (3), so that the diagnostic condition $\varepsilon_v(x) \simeq E_Z(x)$ produces two flank proxy crossings around the pocket. (Under an alternative bookkeeping convention, the window is evaluated with a stray-field-only Zeeman energy, which shifts the resonance windows toward the pocket bottom and, as we show in Sec. V, moves the atlas statistics substantially. This convention sensitivity is itself part of the model risk we report.)

A, the *longitudinal-gradient* profile, is the baseline, motivated by the spatial Zeeman-energy variation the micromagnet imprints. A_{pocket} , the *pocket-localized gradient*, weights that same gradient coupling by a Gaussian envelope centred on the low- ε_v pocket. B_z and B_x , the *longitudinal-* and *transverse resonance-localized* profiles, are resonance-window ansätze motivated by the spin-valley hot spot^{52,55}; their Gaussian windows are centred on the input-detuning condition $\delta(x) \equiv \varepsilon_v(x) - E_Z(x) \simeq 0$, with no claim to a microscopic line shape. They use two alternative choices of magnetic-gradient component: the longitudinal gradient $\partial_x B_z$ sets the local Zeeman-splitting variation (the dephasing gradient^{2,43}), whereas the transverse gradient $\partial_x B_x$ is the component that drives spin flips in slanting-field EDSR^{54,56,57}. Neither choice is prescribed for a shuttled dot by the literature, which is precisely the model uncertainty probed here.

None of the four profiles should be interpreted as a literature-standard or microscopically derived spin-valley coupling; the Gaussian window $e^{-\delta^2/(2\sigma_E^2)}$ in particular is a phenomenological localization envelope on the

coupling amplitude, not a line shape for the enhancement itself. The envelope is bounded by unity and equals unity at $\delta = 0$, so B_z and B_x are the A profile *suppressed away from* the proxy crossing rather than amplified at it; the enhancement near $\delta = 0$ is produced by the four-level dynamics, as the following paragraph sets out. These kernels quantify how strongly the conclusions depend on plausible, non-unique modelling choices.

One might ask whether the gradient-only kernels A and A_{pocket} are physically admissible, given the known spin-valley enhancement near $\varepsilon_v \approx E_Z$. In the diabatic basis of the Hamiltonian above, the operator $\lambda_{sv}(x) \sigma_x \otimes \tau_x$ couples a pair of states whose uncoupled energy difference is $\varepsilon_v(x) - E_Z(x)$, taken in the diagonal part of the diabatic Hamiltonian before Δ_v , B_x and λ_{sv} are diagonalized. The dressed condition is $E_{\text{gap}} \simeq E_{\text{spin}}$, with $E_{\text{spin}}(x) = g\mu_B \sqrt{[B_{\text{ext}} + B_z(x)]^2 + B_x(x)^2}$; we use the diabatic one as a proxy for it, hence “proxy crossings” throughout. A nonzero off-diagonal coupling at a flank crossing therefore produces an avoided crossing and resonantly enhanced spin-valley mixing through the four-level dynamics, even when $\lambda_{sv}(x)$ contains no explicit resonance window^{52,55}.

For the profiles considered here, A and A_{pocket} remain nonzero at the relevant flank crossings. The window in B_z/B_x tests the distinct, stronger phenomenological assumption that the effective off-diagonal coupling amplitude is itself localized near resonance, thereby adding profile localization to the dynamical level-crossing enhancement. Microscopic spin-valley matrix elements depend on spin-orbit and field-gradient matrix elements between valley-dependent envelope states and need not take the particular Gaussian form in $\varepsilon_v - E_Z$ used here^{36,52}. The A-type kernels should therefore be read as diagnostic detuning-independent profiles, not as uniquely derived microscopic matrix elements. Neither family is a null straw man; together they bracket complementary assumptions about the spatial structure of the same off-diagonal coupling.

Figure 2 shows the four profiles for a representative pocket: the gradient-only model A is delocalized over the whole period, A_{pocket} concentrates the gradient coupling near the pocket, and the resonance models B_z/B_x are sharply peaked where $\varepsilon_v(x)$ crosses $E_Z(x)$. An important subtlety arises because all three of A_{pocket} , B_z , and B_x *multiply the field gradient* $|\partial_x B_z|$ (or $|\partial_x B_x|$). The two gradient components are set by the array alone, not by the pocket. By symmetry B_z is extremal and B_x vanishes at the array-symmetry point $x = 0$, so $|\partial_x B_z|$ has a node there; for the fitted baseline field $|\partial_x B_x|$ also attains its maximum at that point. The centred-pocket configuration places $x_c = 0$ there as well, so the node-pocket alignment follows from that placement; it is not a generic property of a centred valley pocket. Consequently $\text{A}_{\text{pocket}} = |\partial_x B_z| \times \text{Gaussian}$ is *zero at the pocket centre* and forms two symmetric peaks on the flanks. A_{pocket} is thus “pocket-localized” in its Gaussian envelope but inherits the gradient’s flank structure; only when the

pocket sits off-centre (where the gradient no longer vanishes at the well bottom) does its weight become strongly asymmetric and single-flank dominated. The resonance windows (B_z , B_x) activate symmetrically on both flanks for the centred pocket as well; the profiles therefore differ not in which side of the pocket they occupy but in their *shape, width, phase relative to the driving gradient, and amplitude*. B_x is built from $\partial_x B_x$ in place of $\partial_x B_z$, and the two gradient components are approximately phase shifted. Its weight therefore peaks slightly outside the crossing, where the $\partial_x B_z$ -based kernels peak slightly inside it. This offset is consistent with the distinct thresholded-classification behaviour B_x shows in the atlas. A left/right asymmetry appears only for an off-centred pocket, where the two flank crossings are inequivalent. The centred-pocket profiles used here are two-sided and symmetric, so any single-sided “opposite side” reading is an argmax tie-break artifact of exactly degenerate twin peaks.

Δ_v and σ_E are toy parameters of the model, not measured constants. The geometric and velocity baselines are representative of reported Si/SiGe shuttling devices.

The observables target the two loss channels that the literature identifies as central to Si/SiGe shuttling: valley excitation (leakage) and valley-induced spin dephasing^{21,29,37,38}. We quantify them as the change in valley-leakage probability ΔP_v and the change in dephasing phase variance $\Delta\chi_\phi$ induced by the spin-valley coupling. The electron enters on the *lower* diabatic valley branch, and P_v is the population of the upper branch at the end of the trajectory; with $\Delta_v \neq 0$ the diabatic branches coincide with the instantaneous valley eigenstates only at the endpoints, so the diabatic statement is the operative one. The two ask different questions: ΔP_v asks how much charge noise changes valley leakage, and $\Delta\chi_\phi$ asks how much the valley sector changes charge-noise dephasing. They are two cross-effects, not two components of one total error, and each therefore needs its own reference. We use three nested models: a spin-only reference with charge noise but no valley sector (M1), a valley-aware reference without charge noise (M1V), and the full model with both charge noise and valley coupling (M2). The two responses use *different* references by design, each separating one cross-effect: $\Delta P_v = P_v^{M2} - P_v^{M1V}$ is the effect of charge noise on valley leakage, and $\Delta\chi_\phi = \chi^{M2} - \chi^{M1}$ the effect of the valley sector on charge-noise dephasing. The dephasing is computed within a filter-function description of $1/f$ charge noise^{18,43,58}. Thus $\mathcal{R} = (\Delta P_v, \Delta\chi_\phi)$ is not a decomposition of a single infidelity but a pair of cross-effect measures. ΔP_v and $\Delta\chi_\phi$ are *baseline-relative response diagnostics*, not absolute error rates. Absolute performance would be read from the full-model values P_v^{M2} and χ_ϕ^{M2} (or a derived phase-error proxy $p_\phi = (1 - e^{-\chi_\phi/2})/2$), which we do not optimize here.

Under the standard assumption that each component is monotonic in the infidelity of its channel (larger leakage and larger dephasing both reduce fidelity), we clas-

Diabatic valley detuning and proxy spin-valley resonance

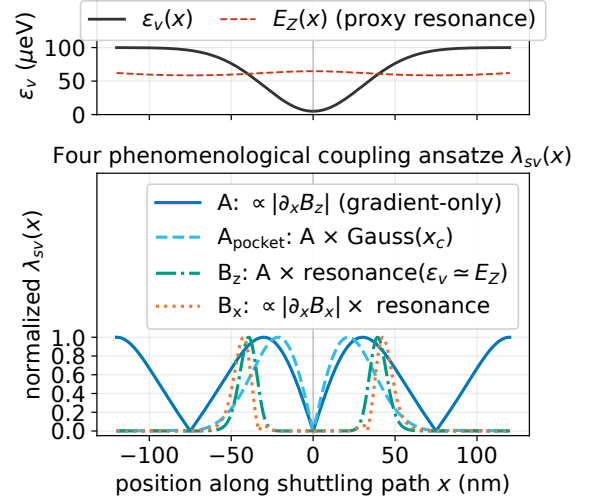


FIG. 2. Diabatic valley-detuning landscape $\epsilon_v(x)$ with a pocket (top) and the four phenomenological coupling ansätze $\lambda_{sv}(x)$ (bottom), normalized to unit peak. All three of A_{pocket} , B_z , and B_x form symmetric twin peaks on the pocket flanks, but for two different reasons. A_{pocket} and B_z multiply the longitudinal gradient $|\partial_x B_z|$, whose node at the array-symmetry point coincides with the centred pocket bottom by placement, so A_{pocket} is zero at the pocket centre despite its “pocket-localized” Gaussian envelope. B_x multiplies $|\partial_x B_x|$, which is instead maximal at the pocket centre; its twin-peak topology therefore comes from the resonance window, not from a central gradient node, since B_z and B_x activate only where $\epsilon_v(x) \approx E_Z(x)$ in the total-field convention. The two gradient components are approximately phase shifted, so at the crossing $|\partial_x B_z|$ is already past its maximum while $|\partial_x B_x|$ is rising out of its node; B_z therefore peaks slightly inside the crossing and B_x slightly outside it. That offset is how the choice of gradient component enters; the two families occupy the same flanks. The top panel plots the diabatic input detuning $\epsilon_v(x)$ of Eq. (1); “proxy resonance” denotes the window $\epsilon_v \simeq E_Z$. For shape comparison only, each displayed curve is independently normalized to unit peak; the adopted prefactor convention retains the unequal final amplitudes in the simulations.

sify each condition by the sign of each *above-threshold* response. The effect-size floors are $|\Delta P_v| \geq 10^{-4}$ and $|\Delta\chi_\phi| \geq 10^{-3}$. A condition clearing both is *robust* if both improve ($\Delta P_v < 0$ and $\Delta\chi_\phi < 0$), *valley-trade* or *spin-trade* if the signs oppose, and *both-worsen* if both worsen. The realization count n_{real} introduced above counts Monte Carlo repetitions of the simulated noise, not experimental repetitions, and is the column name used in the released data tables and scripts. These floors are operational criteria that set a below-threshold zone on channel-appropriate response scales. They come from neither an external fidelity standard nor a statistical-significance calculation, and the decade offset between 10^{-4} and 10^{-3} follows the decade separating the natural scales of the two channels. Sensitivity to shared below-

threshold assignments is examined in the Supplementary Material, and the stability of individual classifications is tested by the seed-stable candidate screen of Sec. V. When only one channel is above threshold, the condition takes a separate single-channel category, which keeps one-channel cases out of the *both-worsen* class. The classification does not compare the two channels' magnitudes against each other: each channel is judged against its own floor, and the category then uses the sign of any above-threshold response. The rank correlation instead uses the response magnitude $|\mathcal{R}| = \sqrt{\Delta P_v^2 + \Delta \chi_\phi^2}$, whose ordering is preserved under uniform rescaling but does depend on the relative weighting of the two channels (we report this dependence as a robustness check).

A co-design decision reads off the direction of the channel responses (whether each channel improves or worsens), not only their magnitude. We therefore treat the four two-channel quadrants as the decision-facing sign diagnostic and the ranking as a complementary magnitude-ordering summary. Both remain informative; they answer different questions and carry different degrees of model dependence.

IV. METHODS

A. Condition grid and reference geometry

We use two pocket placements: $x_c = 0$ for the centred case, aligned with the array-symmetry point, and $x_c = 25$ nm for the off-centred edge case. On top of each we scan nine geometries – the baseline plus one additive + and – step in each of four parameters, **period** (± 30 nm), **depth** (± 10 nm), **half_x** (± 5 nm) and **half_z** (± 3 nm), i.e. $1 + 4 \times 2 = 9$ – together with three velocities and two coupling strengths λ . Here **depth** is the signed vertical coordinate of the quantum well relative to the magnet plane, -50 nm at baseline; a + step moves the well toward the magnet array, reducing the magnet–well separation and strengthening the stray field. The grid holds $2 \times 9 \times 3 \times 2 = 108$ conditions, each evaluated under all four ansätze, giving 432 ansatz-condition cells. The two coupling strengths are $\lambda = 0.5$ and $1.0 \mu\text{eV}$, so the grid contains six (v, λ) families.

Each condition yields the response pair $\mathcal{R} = (\Delta P_v, \Delta \chi_\phi)$ of Sec. III and, from it, a thresholded label drawn from nine categories: the four two-channel quadrants (*robust*, *valley-trade*, *spin-trade*, *both-worsen*), four single-channel categories, and *below-threshold*. We reserve “quadrant” for the four two-channel labels and call exact agreement of the nine-category label *classification agreement*. The label is an exact bijection of the two channels' three-state descriptions (improve, inactive, worsen), a property the channel-resolved diagnostics below rely on. The effect-size floors $|\Delta P_v| \geq 10^{-4}$ and $|\Delta \chi_\phi| \geq 10^{-3}$ are operational criteria, not significance cutoffs or Monte-Carlo confidence bounds.

B. Matched cross-ansatz and cross-seed comparison

The primary experiment measures how far the design-facing label depends on the assumed coupling profile, against how far it moves under a change of random sample alone. We evaluate the full grid in five seed blocks. Within a block, all four ansätze are driven by common random numbers, so every cross-ansatz comparison is made on one realization set; across blocks the realization seeds are disjoint. Each block is evaluated at nested realization counts $n_{\text{real}} = 5, 10, 20, 30, 40, 60, 80, 100$, so the smaller counts are prefixes of the larger and the checkpoints lie on one trajectory, not on independent runs.

Two estimands are defined on the nine-category label L_{acb} of condition $c \in \mathcal{C}$ under ansatz $a \in \mathcal{A}$ in block $b \in \mathcal{B}$, with $|\mathcal{C}| = 108$, $|\mathcal{A}| = 4$ and $|\mathcal{B}| = 5$. For each ansatz pair and each block pair,

$$\begin{aligned} d_{aa'} &= \langle 1[L_{acb} \neq L_{a'cb}] \rangle_{c,b}, \\ d_{bb'} &= \langle 1[L_{acb} \neq L_{acb'}] \rangle_{c,a}, \end{aligned} \quad (8)$$

each a plain mean over the cells in which both members of the pair exist. The two estimands are the uniform averages over pairs,

$$D_{\text{ansatz}} = \frac{1}{6} \sum_{a < a'} d_{aa'}, \quad D_{\text{seed}} = \frac{1}{10} \sum_{b < b'} d_{bb'}. \quad (9)$$

Both are reported with all sixteen pairwise values, together with the difference $\Delta_D = D_{\text{ansatz}} - D_{\text{seed}}$ and the ratio $R_D = D_{\text{ansatz}}/D_{\text{seed}}$. Complete separation of the two pairwise distributions would be sufficient for uniform dominance; it is reported as an observation, not used as a gate, and no numerical interpretation threshold is applied.

D_{ansatz} is itself computed from finite samples and retains an ansatz $\times$ realization interaction, so it is not a pure model-form component. D_{seed} is a finite-sampling label-instability benchmark, not an estimator variance: the classifier is a threshold nonlinearity, and the two quantities are compared, not decomposed. $D_{\text{ansatz}} + D_{\text{seed}}$ is not a decomposition of anything.

A shared *below-threshold* assignment counts as agreement, so the dilution is reported, not conditioned away. Three quantities carry it: p_{all} , the fraction of conditions inactive under every ansatz and block; B_{ansatz} and B_{seed} , the fraction of comparisons in which *both* members are below threshold; and the share of the agreement those account for, $B/(1 - D)$. A union-conditioned recomputation, restricted to comparisons in which at least one member of the pair carries a two-channel quadrant label, is reported as a sensitivity analysis only. Intersection conditioning is not used.

C. Channel-resolved diagnostics

The two responses are isolated against different references by construction, so they carry different baselines

and are never pooled. We therefore repeat the comparison on each channel’s three-state description, which satisfies the exact identity

$$D_{\text{joint}} = D_P + D_\chi - D_{\text{both}} \quad (10)$$

for both the ansatz and the seed averages, D_{both} being the fraction of comparisons that differ in both channels. Equation (10) is an inclusion–exclusion relation: the joint estimand combines the two channel disagreements and their overlap. A near-equality of D_{ansatz} and D_{seed} can therefore arise from opposing channel-level contrasts together with the overlap term, and all three contributions have to be read off before the joint value is interpreted.

For the continuous responses we use a per-channel pairwise dispersion $S^{(q)}$, structurally parallel to the label estimand. Writing $R_{acb}^{(q)}$ for channel q of the response and f_q for that channel’s floor,

$$S_{\text{ansatz}}^{(q)} = \left\langle \left[(R_{acb}^{(q)} - R_{a'cb}^{(q)}) / f_q \right]^2 \right\rangle_{a < a', c, b}^{1/2}, \quad (11)$$

and $S_{\text{seed}}^{(q)}$ likewise over block pairs at fixed ansatz. The pairwise form is used because $\sum_{i < j} (x_i - x_j)^2 = m \sum_i (x_i - \bar{x})^2$ holds for any finite set of m values, so the mean pairwise squared difference is exactly twice the sample variance with the $(m - 1)$ divisor and the 4-versus-5 group-size asymmetry cancels. $S^{(q)}$ is therefore exactly $\sqrt{2}$ times a sample standard deviation – an algebraic identity, not a distributional assumption. $S^{(q)}$ is a dispersion of pairwise differences and carries no information about whether the underlying responses clear their floors.

D. Candidate selection and transport

Operating-point candidates are selected on *seed stability alone*, at the (ansatz, condition) level: a pair is a candidate if the condition is labelled robust under that ansatz in all three discovery blocks. Cross-ansatz agreement is never a selection criterion, so the table is non-circular with respect to the comparison it illustrates. It is not statistically independent of it: the same response data enter both. Blocks 1–3 are the discovery set and blocks 4–5 the holdout. Every condition in the discovery union U^{disc} is then evaluated on the holdout blocks under *all four* ansätze, giving a within-ansatz replication count and a transport count over $0 \dots 4$ ansätze. Because U^{disc} is a set and its membership changes with ensemble size, transport is reported as counts and never as a rate. The same screen run at $n_{\text{real}} = 100$ is an *extended confirmation*, not a second holdout: the $n_{\text{real}} = 100$ aggregate contains the $n_{\text{real}} = 30$ realizations as a prefix and reuses blocks 4 and 5, which were already seen at the smaller ensemble.

E. Registration and the ranking diagnostic

The estimands, the candidate-selection rule and the checkpoints $n_{\text{real}} = 5, 10, 20, 30$ with primary endpoint $n_{\text{real}} = 30$ were fixed in writing before any of these data existed. The extension to $n_{\text{real}} = 100$, including its checkpoints, its hard stop and its reporting language, was registered after the $n_{\text{real}} = 30$ result and before the extension data were generated. The channel decomposition of Eq. (10), the family breakdowns and the ranking diagnostic below were added after seeing results and are marked post-hoc where they appear.

The ranking diagnostic is a complementary magnitude-ordering summary: the pairwise Spearman rank correlation of $|\mathcal{R}|$ across conditions, evaluated within each seed block and then averaged over blocks. It weights the two channels differently from the classifier, since the released magnitude is unnormalized while the classifier judges each channel against its own floor. We test that difference directly by recomputing the correlation under floor-normalized and single-channel magnitudes, and under the signed $\Delta\chi_\phi$.

Nothing here is a population-level statistical claim. There are five registered seed blocks; grid cells are paired by common random numbers across geometry, case and ansatz, and are not independent replicates, so they are never resampled as i.i.d. We report no confidence intervals and no significance tests, and every threshold-dependent statement is scoped to these operational floors.

The distinction between the ranking and the classification is central to our conclusions and is illustrated in Fig. 3: two ansätze may order a set of conditions by response *magnitude* almost identically while still assigning many conditions to different categories. Ranking captures how large the response is; the thresholded label records which response signs are resolved above the floors. Within that label set, the four two-channel quadrants capture the sign pattern of the baseline-relative responses, one ingredient of a co-design decision. A final performance decision would also require the absolute full-model leakage and dephasing proxies, which we do not optimize here.

AI-assisted review. Large-language-model tools were used as secondary review aids during post-processing and manuscript preparation: ChatGPT (OpenAI; GPT-5.5 and GPT-5.6 Sol) and Claude (Anthropic; Opus 4.8 and Opus 5). They reviewed scientific reasoning and claim scope, inspected and debugged consumer-side analysis and plotting code, and checked consistency among the registered estimands, the archived analysis outputs, the figures and the manuscript text. The purpose was to reduce implementation and transcription errors and to add a second consistency check. The registered Monte Carlo runs were executed by the author. Every suggested code or text change was reviewed by the author, and every reported numerical value was verified against the archived outputs. The final interpretations and conclusions were

Quadrant interpretation can flip between ansatz even when the magnitude ranking is similar

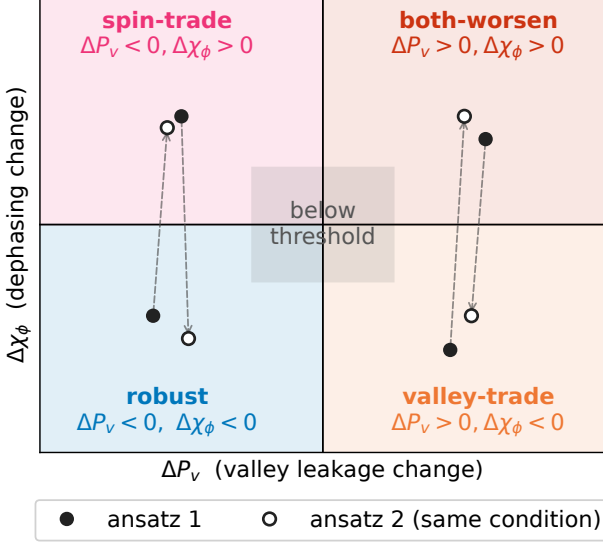


FIG. 3. Quadrant classification of the response $\mathcal{R} = (\Delta P_v, \Delta \chi_\phi)$ into robust, valley-trade, spin-trade, and both-worsen, with a below-threshold zone set by the operational effect-size floors. The panel is a schematic and carries no data axes: the filled and open markers illustrate one operating condition evaluated under two coupling ansätze. Comparable response magnitudes can nevertheless fall into different categories, which is how high magnitude-rank agreement across a set of conditions coexists with classification disagreement on the same set.

determined by the author.

V. RESULTS

a. Unmatched seeds exaggerated the archived model disagreement. The archived analysis drew a separate noise realization for each ansatz, so a cross-ansatz difference there carried a sampling difference as well. Repeating that comparison with the seeding corrected, at matched ensemble size $n_{\text{real}} = 5$, matched seed block and matched convention, raises the mean classification agreement from 35.96% to 56.94% and the mean rank correlation from 0.837 to 0.917. Both endpoints are computed in the archived convention, so the contrast isolates seeding alone. The correction removes 20.99 of the archived 64.04 percentage points of disagreement, about one third; 43.06 points survive matched seeding. It accounts for most of the archived overstatement – and the archived numbers are retained only as a provenance record.

b. Cross-ansatz disagreement is large under matched realizations. At the pre-registered endpoint $n_{\text{real}} = 30$, cross-ansatz disagreement is $D_{\text{ansatz}} = 0.388$ and same-ansatz cross-seed disagreement is $D_{\text{seed}} = 0.385$; the two are comparable there, and both are large. Averaged over conditions, ansatz pairs and seed blocks, about 39% of

matched comparisons assign different design-facing labels. Neither scale dominates at that endpoint, which is a comparison outcome and not a null result.

The aggregate hides a persistent pair structure. Across all eight registered checkpoints the six ansatz pairs fall into three tiers: A/B_z alone at the bottom (0.143–0.180), the three pairs containing B_x in a narrow middle band (0.348–0.407), and A/A_{pocket} together with A_{pocket}/B_z at the top (0.467–0.533). The third pair containing A_{pocket}, A_{pocket}/B_x, sits in the middle band. Between $n_{\text{real}} = 30$ and 100 no pair moves by more than 2.0 points while the seed-pair mean falls from 0.385 to 0.298. The tiers name three specified diagnostic contrasts: among the contrasts examined, the specified pocket localization produces the largest classification change and the specified hot-spot weighting the smallest. The four ansätze form a set of specified contrasts, so this names three of them and not a general ranking of modelling axes. A and B_z remain distinct: they are the most similar pair in the family and still disagree on 16.1% of conditions.

Uniform dominance does not hold. The two pairwise distributions overlap at every checkpoint: at $n_{\text{real}} = 100$ the lowest seed pair, blocks 3 and 5 at 0.169, sits just above the lowest ansatz pair at 0.161, and the two ranges otherwise interleave. As a post-hoc descriptive summary of the same sixty (ansatz-pair, seed-pair) comparisons, 46 favour the ansatz side there, against 33 at $n_{\text{real}} = 30$. Shared inactivity contributes to both sides and is reported, not conditioned away: 13 of 108 conditions are below threshold under every ansatz and block at $n_{\text{real}} = 100$, and shared below-threshold assignments supply 37% of ansatz-pair agreement. Restricting to comparisons in which at least one member carries a two-channel quadrant label, as a sensitivity analysis, leaves the contrast intact and larger, 0.521 against 0.323.

c. The two scales behave differently in ensemble size. Over the eight registered checkpoints D_{seed} decreases from 0.542 to 0.298 with no rebound, while D_{ansatz} stays within a 2.6-point band, 0.369 to 0.395, with no trend [Fig. 1(d)]. The ordering reverses between $n_{\text{real}} = 20$ and 30, and the separation widens thereafter, R_D running from 0.73 to 1.33. Cross-ansatz disagreement thus persists across the ensemble sizes and registered seed blocks examined, whereas cross-seed disagreement decreases over the registered trajectory.

Two limits on reading this. D_{seed} was still falling at the final checkpoint and the extension is hard-stopped there, so no asymptotic or convergence statement is available. And D_{ansatz} is itself computed from finite samples and retains an ansatz×realization interaction, so its flatness is an empirical property of this design, not a property of the model family. At the registered checkpoints up to $n_{\text{real}} = 30$ the two are of comparable size, so a cross-ansatz disagreement reported without the cross-seed benchmark cannot be judged against the label instability a change of random sample produces on its own.

d. The balance is observable dependent. The nine-category label is an exact bijection of the two channels'

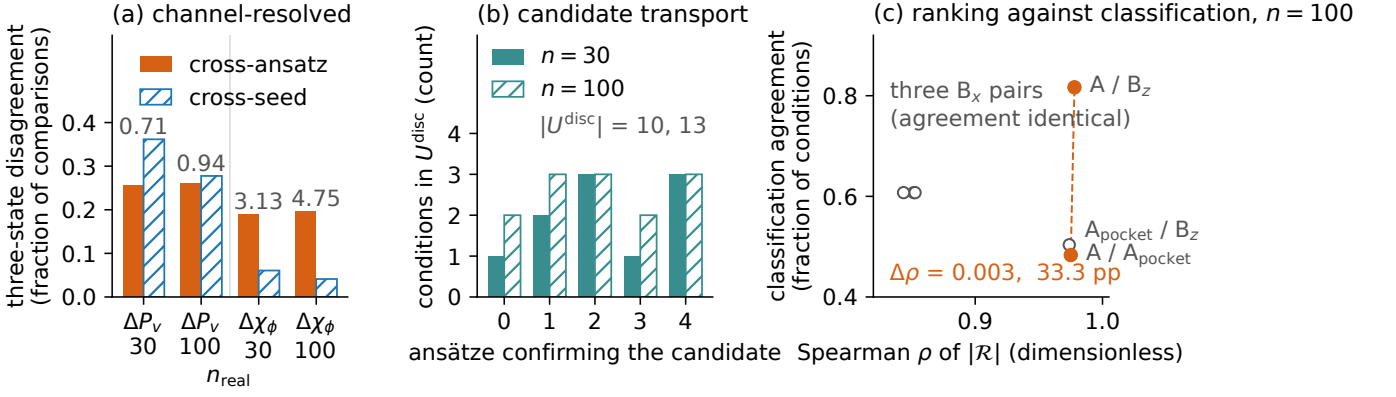


FIG. 4. Channel structure, candidate transport, and the ranking diagnostic. (a) Three-state disagreement per response channel, cross-ansatz against cross-seed, at both endpoints; the number above each pair is their ratio. The valley channel moves from sampling-led at $n_{\text{real}} = 30$ to approximately balanced at 100, while the dephasing channel’s cross-ansatz excess grows. Post-hoc diagnostic. (b) Conditions in the discovery union U^{disc} , binned by how many of the four ansätze confirm them on the holdout blocks; the two columns are $n_{\text{real}} = 30$ (pre-registered) and 100. $|U^{\text{disc}}|$ is 10 at $n_{\text{real}} = 30$ and 13 at 100. The $n_{\text{real}} = 100$ column is an extended confirmation on the same holdout blocks, so the counts are not a before-and-after on one cohort. (c) Pairwise Spearman rank correlation of $|\mathcal{R}|$ against classification agreement, six ansatz pairs at $n_{\text{real}} = 100$. Filled markers joined by the dashed line are the counterexample pair, open markers the other four. A/A_{pocket} and A/B_z differ by 0.011 in rank correlation and 37.2 percentage points in classification agreement. Post-hoc diagnostic.

three-state descriptions, so Eq. (10) holds identically and the joint estimand is an inclusion–exclusion combination of the two channel disagreements and their overlap [Fig. 4(a)]. At $n_{\text{real}} = 30$ the valley channel runs toward sampling, 0.257 against 0.355, and the dephasing channel toward model form, 0.188 against 0.064. The two channel contrasts are opposed, -0.099 and $+0.124$, and the overlap term completes the identity at -0.023 , leaving the joint difference at $+0.002$. The near-equality is the arithmetic outcome of three terms, not one channel prevailing. At $n_{\text{real}} = 100$ the valley channel is approximately balanced, ratio 0.96, while the dephasing ratio rises to 5.60.

The continuous responses agree, and must be stated separately. For $\Delta\chi_\phi$ the cross-ansatz dispersion stays between 40 and 52 in floor units over the whole trajectory and moves from 43.5 to 42.4 between the two endpoints, while the cross-seed dispersion more than halves, 21.8 to 9.8. For ΔP_v both fall steeply – cross-ansatz 28.8 to 23.0 against cross-seed 36.2 to 19.0 – so no ensemble-size-stable cross-ansatz continuous scale is isolated in that channel, and its ratio only reaches 1.22. Neither dispersion is monotone in n_{real} : the seed dispersion rebounds at $n_{\text{real}} = 60$ for ΔP_v and at 10 for $\Delta\chi_\phi$. D_{seed} is the only quantity here that is monotone over all eight registered checkpoints. Which observable a design criterion consumes therefore changes which uncertainty is larger.

e. Ranking and classification are decoupled across the family. This diagnostic and the comparisons in it were added after the registered results were seen; both are post-hoc. Rank correlation is high and narrow across the family, mean 0.913 at $n_{\text{real}} = 100$ with all six pairs between 0.850 and 0.978, while classification agreement spans 0.467 to 0.839. The two do not track each other

[Fig. 4(c)]: A/A_{pocket} and A/B_z differ by 0.011 in rank correlation and by 37.2 percentage points in classification agreement. Chance correction does not rescue the agreement – Cohen’s κ runs from 0.311 to 0.804, and the two lowest values belong to A/A_{pocket} and A_{pocket}/B_z .

The rank metric weights the two channels differently from the classifier, and that was the leading alternative explanation. Recomputing the correlation under a floor-normalized magnitude moves the A/A_{pocket} against A/B_z gap only from 0.011 to 0.023 at $n_{\text{real}} = 100$, and single-channel magnitudes move it less; channel weighting does not account for the decoupling. Among the ranked quantities tested here, the *signed* dephasing response is the only one that separates the counterexample pair: its A/A_{pocket} against A/B_z gap is 0.147, three times the next largest (0.042, the signed leakage response) and an order of magnitude above the released magnitude (0.011). Its six-pair Spearman correlation with classification agreement is $+0.486$ at both $n_{\text{real}} = 30$ and 100, against -0.086 for the released magnitude: the highest of the quantities tested, but not high enough to read the signed response as a substitute ranking. Spin-response sign information is therefore a candidate contributor to the decoupling and not a demonstrated one. It is a post-hoc metric over six non-independent pairs – A/A_{pocket} and A/B_z share A – and the classifier uses floor crossing as well as sign.

f. Seed-stable candidates transport unevenly. Selecting on seed stability alone over the three discovery blocks yields 25 (ansatz, condition) candidate pairs at $n_{\text{real}} = 30$, spread 7/4/6/8 over A , A_{pocket} , B_z and B_x , and a discovery union of ten conditions (Table II). Evaluated on the two holdout blocks, 21 of the 25 replicate within their own ansatz: an operating point that is stable across

TABLE II. Seed-stable candidates at the pre-registered endpoint $n_{\text{real}} = 30$. Selection uses the three discovery blocks and seed stability alone, under a single ansatz; cross-ansatz agreement is never a selection criterion. “discovered by” lists the ansätze under which the condition is robust in all three discovery blocks, and “confirmed” counts the ansätze under which it is robust in both holdout blocks. Geometry labels denote the signed step from the baseline point.

case	geometry	v (m/s)	λ (μeV)	discovered by	confirmed
center	period−	5	1.0	A_p	2/4
edge	depth−	5	1.0	A, A_p, B_z, B_x	4/4
edge	half_z−	5	1.0	A	2/4
edge	period+	5	1.0	A, A_p, B_x	4/4
edge	depth+	10	0.5	A, B_z, B_x	3/4
edge	depth+	10	1.0	A, A_p, B_z, B_x	4/4
edge	baseline	20	1.0	B_z, B_x	1/4
edge	half_z+	20	1.0	A, B_z, B_x	3/4
edge	half_z−	20	1.0	B_x	1/4
edge	period+	20	1.0	A, B_z, B_x	0/4

seeds under one coupling profile generally stays stable under that profile.

Transport across the family is a different matter [Fig. 4(b)]. Of the ten conditions, one is confirmed under no ansatz, two under one, two under two, two under three, and three under all four. The registered $n_{\text{real}} = 100$ companion analysis gives 31 pairs, a union of 13, within-ansatz replication 21 of 31, and transport 2/4/2/2/3. This is an extended confirmation and not an independent replication: the $n_{\text{real}} = 100$ aggregate contains the $n_{\text{real}} = 30$ realizations as a prefix and reuses the same holdout blocks. Of the 25 pairs, 22 are retained, 3 lost and 9 gained, and the two unions are nested. The same three conditions receive four-of-four confirmation at both prefixes. They are grid-local candidates conditional on this diagnostic family, and three conditions do not support a claim about which geometries are robust.

VI. DISCUSSION

Design-facing conclusions in this setting depend on the assumed coupling profile, and the dependence is large under matched realizations. The nine-category thresholded classification is what changes; its four two-channel quadrants encode the decision-facing sign pattern that a co-design choice consumes. Whether a given operating point is labelled robust is contingent on which profile was assumed: three of the ten seed-stable candidates are confirmed under all four ansätze, one under none.

The comparison against cross-seed disagreement is what makes this readable. Showing that four different kernels give four different numbers is true by construction. The informative content is how the two scales behave. Cross-ansatz disagreement stays near 0.39 across a twentyfold change in ensemble size, while cross-seed disagreement falls by nearly half. At the smaller registered

checkpoints the two are of comparable size, so an observed ansatz sensitivity cannot be judged large without the benchmark alongside it, and the judgement changes with the ensemble size at which it is made. It also settles what the earlier archived analysis was measuring: correcting its seeding removes about a third of the disagreement it reported, and the rest survives matched realizations.

Three features of the disagreement structure carry more information than the aggregate. First, ranking and classification come apart: two ansatz pairs differing by 0.003 in rank correlation differ by 33 percentage points in classification agreement, so a family can agree on which conditions respond strongly while disagreeing on which of them improve. The fragile quantity is the classification — the one a design decision consumes. Second, the hierarchy is observable-dependent. Spin dephasing shows a large cross-ansatz excess at both endpoints; valley excitation runs toward sampling at $n_{\text{real}} = 30$ and is approximately balanced at 100. Which loss channel a criterion weights therefore changes which uncertainty is larger. Third, the pairwise disagreements separate into three persistent tiers across all eight checkpoints, so the aggregate conceals a persistent structure among the specified contrasts. These are three diagnostic contrasts within one family, not a factorial design, and they do not identify a general physical axis of disagreement.

The value of the exercise is conditional on a specific fact: no generally validated effective local spin-valley kernel is available for the shuttled-dot and micromagnet regime considered here. A validated microscopic kernel would replace the present diagnostic family as the preferred physical description and would substantially reduce the specific model-form ambiguity studied here. Residual parameter and structural uncertainty could still be assessed with the same comparison. Design decisions today are made under non-unique phenomenological proxies, so the disagreement between those proxies is a real component of design risk, and a single proxy cannot expose it however realistic it looks. Microscopically informed valley landscapes, such as those constrained by alloy-disorder calculations and conveyor-mode measurements^{28,34,37}, can enter the same comparison, and a microscopic coupling can be added as a further member once one is validated for this setting. Whether that narrows the spread is an empirical question.

a. Limitations. All conclusions are scoped to four diagnostic coupling profiles, one fixed minimal spin-valley operator, one fixed charge-noise description, and the periodic micromagnet reference geometry examined here. The four ansätze differ only in their scalar spatial profiles, so operator-level alternatives are outside the present scope. The same $1/f$ spectrum and a single representative, non-device-calibrated displacement-noise amplitude enter both responses, since ΔP_v measures the effect of charge noise on valley leakage and $\Delta\chi_\phi$ the effect of the valley sector on charge-noise dephasing. What is quantified is therefore coupling-profile uncertainty con-

ditional on that noise description; noise-spectrum and amplitude uncertainty are a separate axis and are not scanned.

The thresholded labels are operational classifications under the stated effect-size floors, not confidence-certified assignments, and every threshold-dependent statement is scoped to those floors. Sensitivity to shared below-threshold assignments is examined in the Supplementary Material. Nothing reported here is a population-level statistical claim: there are five registered seed blocks, grid cells are paired by common random numbers and are not independent replicates, and no confidence intervals or significance tests are computed.

The registered trajectory stops at $n_{\text{real}} = 100$, where cross-seed disagreement was still falling, so no asymptotic or convergence statement is available. Cross-ansatz disagreement is itself computed from finite samples and retains an ansatz $\times$ realization interaction; its stability across the trajectory is an empirical property of this design. The $n_{\text{real}} = 100$ candidate layer is an extended confirmation on blocks already seen at $n_{\text{real}} = 30$, not an independent replication. The three conditions confirmed under all four ansätze are grid-local and conditional on this diagnostic family; three conditions do not support a claim about which geometries are robust.

Natural extensions are to measured or finite-device micromagnet fields, to families of noise descriptions, and to microscopically derived couplings and operator structures. We leave these to future work because the purpose here is to isolate coupling-profile dependence against a fixed geometric backdrop.

b. Implications for co-design. Two practical implications follow. First, a claim that a particular operating point is robust should state which coupling profile produced it, and should report whether the point transports across a family. Proposed mitigations such as varying the shuttling velocity, dot shape, or path²⁹ deserve the same treatment. Second, a cross-ansatz comparison should carry a same-ansatz cross-seed benchmark at the same ensemble size, since at the smaller checkpoints examined here the two are of comparable size.

A geometry or trajectory optimum obtained under one assumed coupling profile remains model-conditional. Two routes strengthen it: constraining the effective coupling independently, against microscopic theory or experiment, or demonstrating that the optimum survives across a justified model family. The comparison reported here supplies the second route’s measurement, and says nothing about which route is cheaper.

VII. CONCLUSION

We have compared cross-ansatz and cross-seed variation in a design-facing classification for spin shuttling through a periodic micromagnet field, under common random numbers and across a registered range of Monte-Carlo ensemble sizes. Two results are methodologi-

cal. First, drawing separate noise realizations per model mixes model differences with realization differences: in the archived analysis, correcting the seeding removes about a third of the disagreement it reported, and two thirds survive. In the finite-ensemble paired comparison used here, matched realizations are what isolate ansatz dependence from realization differences; without them an observed cross-ansatz difference mixes the two. Second, the two scales behave differently in ensemble size, so a cross-ansatz disagreement is interpretable only against a cross-seed benchmark measured at the same ensemble size.

The scientific result is that cross-ansatz disagreement in this setting is large, persistent across the registered ensemble sizes and seed blocks, and unevenly distributed. It concentrates in the thresholded classification while magnitude ranking stays high; its excess over cross-seed disagreement is much larger in the dephasing channel than in valley excitation; and it separates the specified contrasts into three persistent tiers. Operating points selected for seed stability replicate within their own ansatz and transport unevenly across the family. Design conclusions drawn from a single phenomenological coupling profile therefore remain model-conditional.

a. Future work. The natural next step is a microscopic effective H_{sv} for this regime, derived from orbital admixture, valley-orbit interaction, and g -factor anisotropy^{35,36,54}. Related gradient, strain, and spin-orbit design questions in Ge hole-spin platforms^{40–42,47,48} can inform the same program. Because the framework treats the coupling profile as an interchangeable module, a microscopic model enters as an additional member and is compared pairwise against the diagnostic family. Residual uncertainty in microscopic input parameters will likely leave a finite degree of model dependence, which the same comparison would measure. A factorial design over gradient component and spatial localization, which the four profiles here cannot support, would test whether the disagreement has an identifiable physical axis.

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AUTHOR DECLARATIONS

Conflict of interest

The author has no conflicts to disclose.

Ethics approval

This work is entirely theoretical and computational. It involved no human participants, animal subjects, or personal data, and required no ethical approval.

Author contributions

Inuk You: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.21588784>, reference number 21588784 (all versions: [10.5281/zenodo.21588003](https://doi.org/10.5281/zenodo.21588003)), and are mirrored at https://github.com/youinuk/spin-valley_model. The archive contains the simulation code, the analysis outputs behind every reported number, and the scripts that regenerate every figure. All quantitative results of the matched comparison are reproducible from those outputs with the registered seed blocks; quantities identified in the text as archival retain the seeding and release provenance of the earlier record.

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